

Construct an Embedded C simulation for a wearable healthcare device with low power and real-time operation.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

void enterSleepMode()
{
    printf("System in Low Power Mode...\n");
}

void generateAlert()
{
    printf("*** ALERT : Abnormal Health Condition Detected **\n");
}

int main()
{
    int heartRate;
    float temperature;
    int spo2;

    srand(time(NULL));

    while(1)
    {
        printf("\n-----\n");
```

```

// Simulated Sensor Values

heartRate = 55 + rand()%70;

temperature = 35.5 + (rand()%50)/10.0;

spo2 = 88 + rand()%12;


printf("Heart Rate : %d bpm\n",heartRate);
printf("Temperature: %.1f C\n",temperature);
printf("SpO2      : %d %%\n",spo2);


// Real-Time Health Check
if(heartRate < 60 || heartRate > 100 ||
   temperature > 38.0 ||
   spo2 < 92)
{
    generateAlert();
}
else
{
    enterSleepMode();
}


// Delay Simulation
for(long i=0;i<200000000;i++);
}

return 0;
}

```

Output:

Heart Rate : 69 bpm
Temperature: 39.7 C
SpO2 : 88 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 115 bpm
Temperature: 40.1 C
SpO2 : 98 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 92 bpm
Temperature: 38.0 C
SpO2 : 96 %
System in Low Power Mode...

Heart Rate : 89 bpm
Temperature: 39.1 C
SpO2 : 98 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 111 bpm
Temperature: 36.2 C
SpO2 : 89 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 99 bpm
Temperature: 36.6 C
SpO2 : 89 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 59 bpm
Temperature: 38.9 C
SpO2 : 92 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 118 bpm
Temperature: 38.9 C
SpO2 : 92 %
** ALERT : Abnormal Health Condition Detected **

Heart Rate : 115 bpm

Temperature: 39.5 C

SpO2 : 88 %

** ALERT : Abnormal Health Condition Detected **

Heart Rate : 78 bpm

Temperature: 36.0 C

SpO2 : 90 %

** ALERT : Abnormal Health Condition Detected **

Heart Rate : 111 bpm

Temperature: 40.0 C

SpO2 : 93 %

** ALERT : Abnormal Health Condition Detected **

Heart Rate : 63 bpm

Temperature: 36.8 C

SpO2 : 95 %

System in Low Power Mode...